A Deep Learning Approach to Mapping Irrigation Using Landsat: IrrMapper U-Net

Thomas Colligan, David Ketchum, Douglas Brinkerhoff, and Marco Maneta

***Abstract*— Accurate maps of irrigation are essential for understanding and managing water resources. We present a new method of mapping irrigation based on an ensemble of convolutional neural networks that use reflectance information from Landsat imagery to classify irrigated pixels. The methodology does not rely on extensive feature engineering and does not condition the classification with land-use information from existing geospatial datasets. The ensemble does not need exhaustive hyperparameter tuning, and the analysis pipeline is lightweight enough to be implemented on a personal computer. Furthermore, the proposed methodology provides an estimate of the uncertainty associated with the classification. We evaluated our methodology and the resulting irrigation maps using a highly accurate novel spatially explicit ground-truth dataset, using county-scale United States Department of Agriculture (USDA) surveys of irrigation extent, and using cadastral surveys. We demonstrate the accuracy of the method by mapping irrigation over the state of Montana from 2000 to 2019. We found that our method outperforms other methods that use satellite remote-sensing information in terms of overall accuracy (OA) and precision. We found that our method agrees better statewide with the USDA National Agricultural Statistics Survey estimates of irrigated areas compared to other methods and has far fewer errors of commission in rainfed agriculture areas. This methodology has the potential to be applied across the entire United States and for the complete Landsat record.**

***Index Terms*— Irrigation, Landsat, neural networks.**

I. Introduction

IRRIGATED cropland is responsible for approximately 40% of the world's crop production, despite only comprising 20% of all cropland by area [1]. Irrigation already makes up about 85%–90% of all consumptive freshwater use globally [2]. The Food and Agriculture Organization of the United Nations (FAO) forecasts that by 2030, the world will have increased irrigated land by 34% [1]. Despite the positive impact irrigation has on crop production, it is associated with a range of environmental impacts ranging from aquifer and surface water depletion to soil salinization. The reliance of agriculture on irrigation makes spatially explicit knowledge of when irrigation is occurring and how it is changing over time essential to understanding and managing water resources. Maps of irrigation can also help inform climate and earth systems models, for example, by refining the crop evapotranspiration component of land surface models, and by helping establish more realistic estimates of agricultural water diversions from surface and groundwater sources. Most large-scale maps of irrigation have been produced at low resolutions and/or rely on remote-sensing data combined with national or subnational statistics [3], [4], [6]–[9], [33]. Agricultural statistics collected by governments or non-governmental organization (NGO) often do not capture high-resolution temporal trends in irrigation due to labor-intensive collection of data that often contains biases, especially if there are political or other incentives for over-reporting or under-reporting irrigation.

Methods based solely on census data are limited in resolution by the administrative boundaries used to aggregate and report the data. For example, the United States' National Agricultural Statistics Survey (NASS), which reports irrigated area and irrigated production by county in the U.S., is compiled yearly by self-reported survey, with more accurate statistics of irrigation extents compiled every five years for the agricultural census. The statistics are generally regarded to be an accurate characterization of crop type and irrigation status for lands across the U.S., but could be biased due to incentives to over- or under-report [2], [33]. The self-reporting and spatially aggregated nature of these datasets do not permit an accurate assessment of how irrigation extents have changed over time and space. In addition, NASS excludes irrigated areas that produce less than $1000 [10] of production value, adding further biases to the surveys by discarding small-scale but numerous agricultural operations. Definitions of irrigation may also depend on the criteria of the organization collecting the data and may often include land equipped for irrigation, not actually irrigated area [33]. Irrigated area and area equipped for irrigation may be significantly different in any given year due to fluctuations in crop prices or rotation [11]. The spatial resolution (ranging from 500 m to 5 arcsec) of many existing maps of irrigation of the globe and the continental United States (CONUS) makes the identification of small or partially irrigated areas difficult. The low resolution of previous maps of irrigation on large scales is due to the extensive computing resources required to process satellite data and the effort required to gather ground-truth samples.

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However, the increasing ability of researchers to work with large datasets has made mapping irrigation at high resolution possible.

Google earth engine (GEE) [12] has aggregated remote-sensing data in one place and provided free access to their computing infrastructure, helping researchers increase the resolution and scale of remote-sensing mapping approaches. The introduction of GEE to the remote-sensing community has made mapping irrigation at high resolution on large temporal and spatial scales possible without relying on local computing power. This shift in technology has led to multiple efforts at mapping irrigation using Landsat data on large scales and high resolutions [2], [9], [13], [14]. These efforts all rely on random forest (RF) models and novel ground-truth datasets to map irrigation across regions ranging from the High Plains Aquifer to the entire CONUS. RFs are common in the remote-sensing community for their flexibility, interpretability, ease of use, and their ability to handle complicated high-dimensional data. Deines *et al.* [2], Xie *et al.* [9], Deines *et al.* [13], and Ketchum *et al.* [14] are the only studies that describe high-resolution (30 m) irrigation datasets developed at regional scales over long periods in the United States. Landsat-based irrigation dataset (LANID) [9] is a Landsat-based irrigation map that uses a semiautomatic method of generating training data to train RF models that classify irrigation across CONUS at 30 m. The semiautomatic generation of training data uses estimates of the irrigated area by county from the Moderate Resolution Imaging Spectroradiometer (MODIS) Irrigated Agriculture Dataset for the United States (MIrAD-US) [3] (a 250-m irrigation map that matches normalized difference vegetation index (NDVI)-derived estimates of irrigation area directly with NASS statistics by county) to threshold maps of satellite-derived spectral indices until the area reported by the maps and MIrAD match. The thresholded maps serve as areas in which [9] sample points to train RF models, which are used to classify land as irrigated or nonirrigated. The LANID method is based on county-level statistics through the reliance on training data generated by matching MIrAD-US. IrrMapper [14] and Annual Irrigation Maps–High Plains Aquifer (AIM–HPA) [2], take a different approach and train their RF models with hand-labeled training points, producing estimates of irrigated land independent of NASS. IrrMapper covers the western United States and AIM-HPA extends across the High Plains Aquifer. Both LANID and AIM-HPA mask their classification product to an ancillary land cover map such as the cropland data layer (CDL). IrrMapper (and our method) removes this postprocessing step, instead of using the CDL as an input feature. The RF methods above (LANID, IrrMapper, and AIM-HPA) all report overall accuracy (OA) greater than 90% and agree well with NASS statistics.

Part of RF's widespread use is due to its ease of use and free implementation in GEE. However, the state of the art in image segmentation (of which irrigation mapping is a subset) is achieved through the application of deep convolutional neural networks (CNNs). CNNs can match and even exceed human performance in various computer vision tasks. CNNs incorporate learned representations of images on many scales when making predictions, picking up patterns and contextual information. This is especially important in irrigation mapping, where the difference between a nonirrigated field and irrigated field may not be solely in spectral information but in the shape of the field or surrounding context. RF models, in contrast, have to rely on handcrafted features that capture spatial context or otherwise discard spatial information in their classification. Another advantage of CNNs is that they can use dense pixel-wise labels, while RF models have to pick and choose which points they use for training. CNNs have been used in many domains of remote sensing, from crop-type mapping and land cover classification [15], [16], to change detection [17], [18]. Despite their proven utility in remote-sensing applications, there are only a few examples of CNNs directly applied to producing maps of irrigation, especially at large scales. The main reason for this deficit is the limited availability of training data [13] and high computational costs. Training deep neural networks effectively requires thousands of training, testing, and validation samples as well as specialized computer hardware [e.g., graphical processing units (GPUs)]. Evaluating these models over a large geographic region at high resolution can also be cost-prohibitive, as CNNs are often computationally expensive. Despite the disadvantages, the accuracy of CNNs routinely surpasses that of RF-based approaches because of their ability to learn explanatory features directly from data. The only use of deep learning to produce maps of irrigation to our knowledge at the time of writing is the work by Zhang *et al.* [19] and Saraiva *et al.* [20]. Both methods use CNNs to map center pivot irrigation patterns over the state of Colorado and the Cerrado Biome of Brazil, respectively. These methods both relied on intensive hand-labeling of center pivot irrigation structures in satellite images. Both achieved an OA of greater than 99%, and limited their study area to a relatively small region in time and space (20 000 km^2, 1986 and 2000, and 37 000 km^2, 2017, 2018, and 2019, respectively).

In this article, we present a neural network model and implementation framework that is easy to use and achieves better results than current approaches to mapping irrigation at large spatial and temporal scales. We use novel ground-truth data (a subset of the data used in [14]) to train an ensemble of neural networks to label Landsat pixels as irrigated or nonirrigated. We demonstrate the methodology for the state of Montana (MT) for the years 2000–2019. Unlike other products, our method produces pixel-wise estimations of classification uncertainty associated with each irrigation map. We conduct a spatially explicit, pixel-wise comparison between methods (IrrMapper, U-Net ensemble, and LANID) using the novel ground-truth data. Our method outperforms other methods in OA and precision (also known as errors of commission), despite limited feature engineering, and the omission of a crop mask like the CDL to condition the classification to cropland pixels. Omitting this information permits the application of the method to periods and regions not covered by the crop mask. Unlike other products, our method does not require a preprocessing step to mask clouds or missing data because the ensemble of neural networks learns to fill data gaps. To assess our methodology, we compare our estimations to NASS statistics and final land unit (FLU) [21] classification data from MT. We find better agreement for most years between the total irrigated area in MT calculated with our approach and NASS data compared to existing maps

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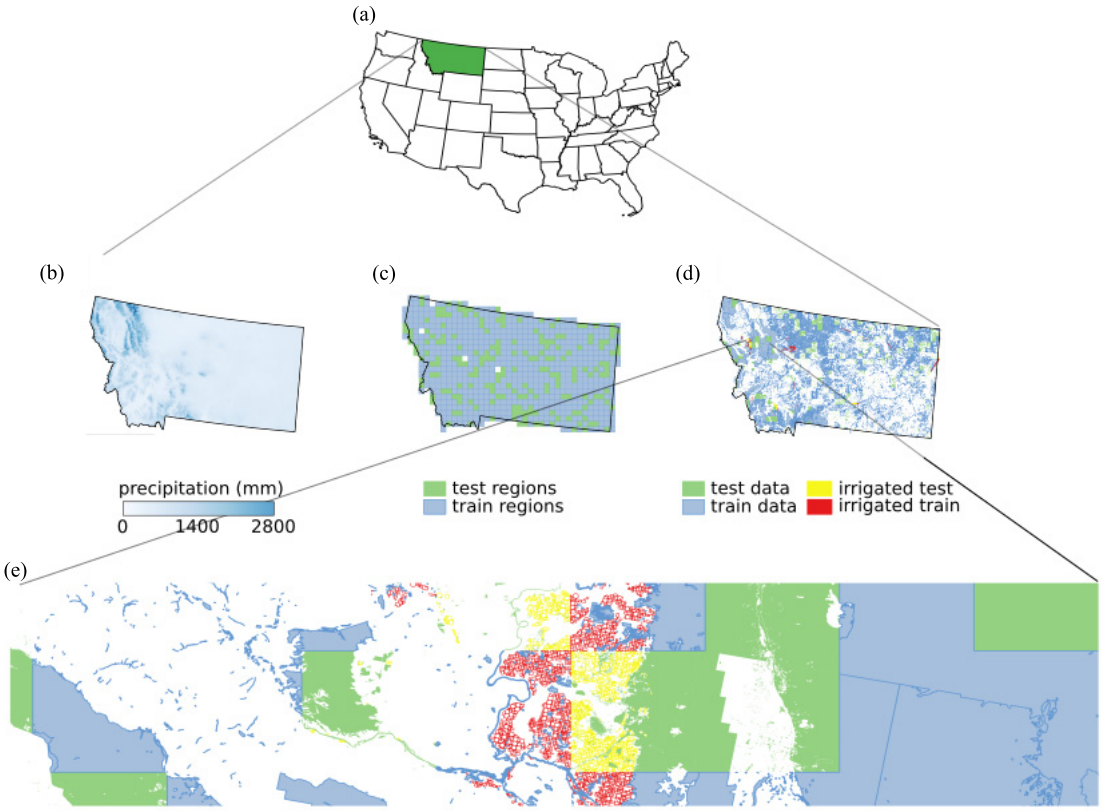


Fig. 1. Study area and data used in this analysis. (a) United States, with MT highlighted in green. (b) Mean annual precipitation for the years 2000-2019. (c) Test and train regions spread over the state of MT. (d) Training and test polygons. A zoomed-in view of the training and test polygons is shown in subfigure (e). Irrigated training polygons are in red and test polygons are in yellow. For ease of visualization, the test/train unirrigated and uncultivated classes are aggregated into one class (test shown in green, train shown in blue in (c), (d), and (e).

of irrigation. We also report on the qualitative performance of the U-Net ensemble, LANID, and IrrMapper, describing the successes and failures of each product.

II. Methods

A. Spatial and Temporal Study Region

The development and demonstration of the methodology focus on the state of MT, in the United States northern Rockies, with an area of ~380 000 km^2 (Fig. 1). The state of MT has a marked longitudinal topographic and climatologic gradient, and a mixture of rainfed and irrigated agriculture and ranching that is representative of the conditions of the U.S. intermountain west and makes it a good site to test our methodology. The western third of the state is dominated by complex topography, steep gradients, and relatively smaller properties. The eastern portion transitions to the Northern Great Plains region, with gentler topography and larger ranching operations. West of the Continental Divide, MT is generally wetter and more temperate than the area east of it due to the influence of Pacific moisture. The climate of the eastern portion has stronger continental features and is drier and windier during the growing season [22]. MT's mean annual precipitation for 2000–2019 is shown in part (b) of Fig. 1. Surface water provides 99% of irrigation water withdrawals in the state of MT [23] and comprises ~68% of consumptive water use [22] in the state. We applied our methodology in this region and produced irrigation maps for the years 2000–2019.

B. Hand-Created Set of Labels

We used a subset of the training dataset used by Ketchum *et al.* [14]. The dataset consists of vector polygons delimiting sample regions for three classes: irrigated, unirrigated, and uncultivated land. All field labels were created through manual interpretation of satellite images, National Agricultural Imagery Program (NAIP) imagery, and field surveys. Unirrigated labels contain predominately dryland (rainfed) agriculture and include some fallow fields. The uncultivated labels contain land which is not used for crop cultivation, like rivers, lakes, wetlands, forests, and prairies. Irrigated and fallow field labels were used for 2003, 2008–2013, and 2015 to represent irrigated land during a range of water availability, using yearly precipitation as a proxy.[1] Areas labeled as unirrigated or uncultivated were assumed to stay the same class over the years studied. Overall, the models were trained and validated with data for the years 2003, 2008–2013, and 2015, with spatially distinct training and validation sets.

[1]Using the National Atmospheric and Oceanic Administration (NOAA) Climate At A Glance tool, annual precipitation was plotted for 2000–2019 (https://www.ncdc.noaa.gov/cag/statewide/time-series/24/pcp/ann/1/2000-2021?base_prd=true&begbaseyear=1901&endbaseyear=2000). Years were selected in this time range to build training data if they had diverse amounts of precipitation.

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To create spatially independent training and validation data, we overlaid a grid onto the state of MT. Each grid square was $23\,040^2$ m^2, with a total number of 776 squares. The squares were then split into sets 582 and 194 squares each (80, 20% splits, part (c) in Fig. 1). We shuffled the ordering of the squares and then labeled the first 80% as training and the rest test. The hand-created set of labels described above was then split into training and validation sets based on the grid subsets. Specifically, we used Quantum Geographic Information System (QGIS) "clip" tool to extract labels that intersected the training and test grid. This block dataset partitioning ensures that the method is not simply learning to fill in gaps, as might occur if we simply randomly sampled individual pixels.

C. Features

We used Landsat Tier 1 Surface Reflectance data from Landsat missions 5, 7, and 8, acquired through GEE. The Landsat mission is the longest continuous record of satellite images and minimal effort is required to standardize across missions. Landsat images of the earth are also captured frequently, with approximately eight days between subsequent observations across all satellites. The bands used for analysis were blue, green, red, near-infrared response (NIR), shortwave infrared (SWIR 1), and SWIR 2. We did not perform any cross-calibration between Landsat sensors, nor did we mask clouds or cloud shadows.

To standardize data across years, we took temporal means of images captured over a 32-day period, inspired by Allred *et al.* [24]. All available Landsat images starting on May 1 and extending for 192 days (six 32-day periods) were collected and sorted by time using GEE. For each nonoverlapping 32-day segment within the 192 days, the temporal mean of all images collected during that segment was computed, and the resulting mean image was stored and used as a feature of the model. The final features that were ingested into the model were six temporal mean images (each comprised of six bands: blue, green, red, NIR, SWIR 1, and SWIR 2 for a total of 36 bands) sorted by time.

We treat each band as a separate feature without explicit temporal information. The 32-day time period was chosen because it represented a tradeoff between temporal resolution and the number of input features. Taking the mean over a longer time period could suppress important information for determining irrigation. A shorter aggregation period would cause the number of input features to rise, increasing computational and storage needs. The training and validation data are not processed in any way to compensate for missing data due to the Landsat-7 scan-line failure. The failure occurred on May 31, 2003, so the scan-line failure was present in all of the training and validation data. Missing data due to the failure was noticeable in the data upon inspection. The machine learning (ML) algorithm has to learn a representation of irrigation that ignores scan line artifacts.

D. Model Description

The CNN used to predict irrigation was a variant of the U-Net model [25], with approximately 4× fewer parameters than the original. The U-Net architecture was chosen because it is a tried-and-true method for image segmentation and because it is fully CNN (FCNN). FCNNs are noted for their fast inference time compared to pixel-wise methods [26]. Reducing the number of parameters reduced the time required for training and inference. See the Appendix for more details on CNNs and the model architecture used in this study.

E. Training and Ensemble Details

We trained ten models on random subsets of the training data, as model ensembles are shown to do better than their constituents and can provide a measure of predictive uncertainty [27]. Each model was trained for 100 000 gradient descent steps (batch size 25) with the Adam [28] optimizer with default parameters. We used random majority undersampling [29] to roughly balance examples from each class per batch. The initial learning rate was 0.001 and was step decayed every 25 000 steps by 60%. Each model took around 24 h to train on a single NVIDIA GTX 1070 GPU.

F. Ensemble Uncertainty

Trained models were used to predict irrigation status across the study region. Each model outputs a real number in the range 0–1 for each pixel, indicating the strength of the model's belief that a pixel belongs to a specific class. Each pixel is assigned three numbers, one for each class, that sum to 1. The interquartile range (IQR) of the model predictions for each pixel and each class was used as the uncertainty of the ensemble. Smaller IQR indicates high confidence in the ensemble. Maps of IQR were produced for the entire study region.

G. Map Postprocessing

We used the overlap-tile strategy in [25] to produce seamless predictions over the region of interest for each model. The median prediction of the ten models was used as the final prediction when creating maps of irrigation. The ensemble results are referred to as the "U-Net ensemble" below. To reduce data storage requirements, the output of the models was scaled to the range 0–255 (from 0 to 1) and converted to an unsigned 8-bit integer.

Landsat pixels can contain multiple different land classes. For example, pixels that capture an edge of a field may also contain the land adjacent to the field, which may be a road or a different field. Since there are many roads next to irrigated fields, the models learned to classify pixels at the edges of fields that contain roads as irrigated. Rather than incorporating roads into the set of classes the models had to distinguish, we used human-labeled data to postprocess roads out of model predictions.

We used the rasterio.mask.mask function from the rasterio python library [30] to remove the pixels labeled as roads in the roads shapefile from the final predictions. We used a shapefile containing all MT roads created by the Montana Department

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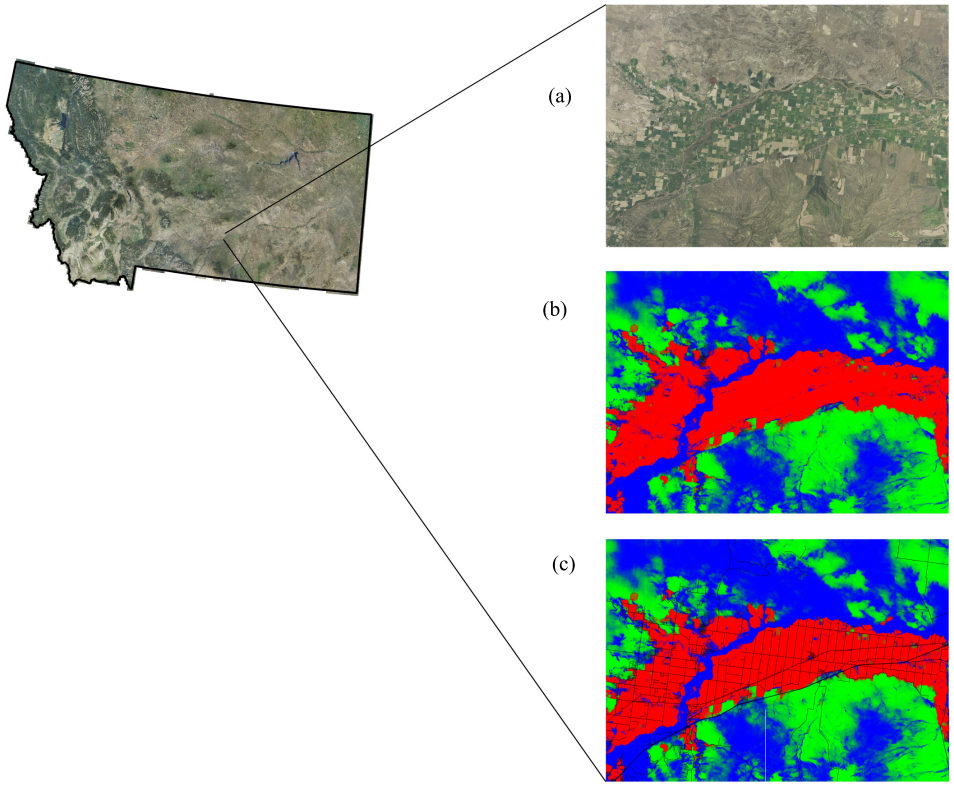


Fig. 2. (a) NAIP imagery of irrigation near Billings, MT (2013). NAIP imagery only used for visualization, not for analysis. (b) Corresponding unmasked model predictions. Red indicates area predicted to be irrigated. (c) Irrigation predictions after masking with roads shapefile.

of Transportation as the mask.[2] An example of the initial predictions and the masked predictions is shown in Fig. 2.

H. Accuracy Analysis and Validation

We conducted a pixel-wise comparison of LANID, IrrMapper, and the U-Net ensemble on the spatially distinct validation set derived from the set in [14]. Since each member of the U-Net ensemble had $4\times$ fewer parameters than the original U-Net, we also compared our method to a single U-Net with the original number of parameters trained on the nonbootstrapped training dataset to show that there was no significant loss in accuracy. The validation data consisted of hundreds of millions of pixels, with $\sim 100\times$ more non-irrigated pixels than irrigated pixels. F1-score was used to quantify errors of omission and commission on irrigated labels. F1-score aggregates the number of true positives (TP), false positives (FP), and false negatives (FN) into one number, defined as $\text{f1} = (\text{TP}/(\text{TP} + (1/2)(\text{FP} + \text{FN})))$. An alternative formulation of F1-score relies on precision and recall (also known as errors of commission and omission, respectively), $\text{f1} = ((2 \times P \times R)/(P + R))$, where $P = (\text{TP}/(\text{TP} + \text{FP}))$, and $R = (\text{TP}/(\text{TP} + \text{FN}))$. We use F1-score, precision, and recall to quantify the performance of the various methods (LANID, IrrMapper-RF, and the U-Net ensemble) on the irrigated class. A pixel was considered to be irrigated if the median U-Net ensemble prediction for irrigation was larger or the same as the median prediction for the other two classes. We also report a three-way confusion matrix between predictions and labels, presented as both pixel counts and counts as a percentage of total labeled pixels (delimited by a "/" character).

To ensure our method was able to generalize in time and space [31], we trained and tested multiple models on spatially and temporally distinct sets of data. We held out three years from the training data (2008, 2013, and 2015), and trained three separate bootstrapped ensembles of models on the held out sets. The three years were chosen randomly. Due to computational constraints, we performed this analysis with an ensemble size of four. F1-score was used to quantify errors of commission and omission on the spatially and temporally distinct test set. We also compare the ensemble of four to a single model randomly chosen from the ensemble to show the benefits of an ensemble approach.

IrrMapper, MIrAD-US, and LANID are the only other high spatial and temporal resolution irrigation maps that cover the state of MT. MIrAD-US is only available for the years 2002, 2007, 2012, and 2017, because it depends on NASS statistics (as such, it agrees very well with NASS predictions of the irrigated area by county). We did not produce pixel-wise assessments of MIrAD-US maps since they are 250 m resolution compared to the 30 m of LANID, IrrMapper, and the U-Net ensemble. IrrMapper covers the range 1986–2019, and we were able to access LANID for 2008–2013. To compare between models, we aggregated the U-Net ensemble classes

[2]Downloaded from https://geoinfo.msl.mt.gov/home/msdi/transportation/

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TABLE I
CONFUSION MATRIX OVER A SPATIAL VALIDATION SET COMPUTED FROM THE ENSEMBLE'S MEDIAN PREDICTION

		Predicted			
		Irrigated	**Unirrigated**	**Uncultivated**	**Recall**
Actual	**Irrigated**	1788615/0.81	184405/0.084	104779/0.048	0.86
	Unirrigated	114903/0.052	104929537/47.8	1212801/0.55	0.99
	Uncultivated	194506/0.089	5334847/24.3	105872279/48.2	0.95
	Precision	0.85	0.95	0.97	

TABLE II
OA, PRECISION, RECALL, AND F1 SCORE FOR THREE DIFFERENT IRRIGATION PRODUCTS

		Overall accuracy	**P**	**R**	**F1**
Method	IrrMapper RF	0.993	0.63	0.88	0.73
	LANID RF	0.988	0.49	0.84	0.61
	U-Net Ensemble (Ours)	0.997	0.87	0.88	0.88
	Full U-Net	0.963	0.90	0.80	0.84

TABLE III
F1-SCORES FOR MEDIAN ENSEMBLE U-NET RESULTS TRAINED ON SPATIAL AND TEMPORALLY DISTINCT VALIDATION SET, ORGANIZED BY HELD-OUT YEAR

	2008	**2013**	**2015**
single model	0.83	0.86	0.81
ensemble of 4	0.87	0.87	0.88

unirrigated and uncultivated into a single class (nonirrigated), as the other products only assign a binary indicator to each pixel, and evaluated each model's performance on the validation split.

To assess the difference between model predictions of irrigated land and agricultural statistics data, we compared U-Net ensemble results to two different estimates of irrigated areas: NASS and FLU. NASS statistics were available for the years 2002, 2007, 2012, and 2017, as shown in Fig. 3. FLU is a vector dataset produced by the Montana Department of Revenue for agricultural property valuation that classifies private agricultural land into one of six uses: fallow, hay, grazing, irrigated, continuously cropped, and forest. FLU estimates of the irrigated area were aggregated by county and were available for the years 2009, 2011, 2013, 2015, 2017, 2018, and 2019. Agreement between the U-Net ensemble, FLU, and NASS was assessed with slope and r^2.

III. RESULTS AND DISCUSSION

The confusion matrix in Table I shows the performance of the median model on the spatial validation set. Over all years considered, the U-Net ensemble had an F1-score for the irrigated class of 0.86 and 0.97 for both the uncultivated and the unirrigated classes. There were different numbers of irrigated labels in different years. The year 2003, for example, only had $\sim 3 \times 10^4$ irrigated pixels and over 2.7×10^7 nonirrigated pixels. The F1-score for 2003 was only 0.4, but for years with less imbalanced distributions (like 2013, with $\sim 3.5 \times 10^5$ irrigated pixels) the F1-score jumped to around 0.9. Despite the fact that the F1-score was variable depending on the number of pixels labeled as irrigated per year, the model predictions looked similar throughout the time period evaluated.

A quantitative comparison of the performance of the different mapping techniques is shown in Table II. Metrics are only presented for the irrigated class. The U-Net ensemble outperforms existing maps of irrigation in Table II. There is a slight discrepancy between F1 scores in Tables I and II because labels from the years 2003 and 2015 are omitted in Table II to keep the comparison between methods based on the intersection of the set of years where each product (IrrMapper-RF, LANID, U-Net ensemble) is available. The full U-Net in II has a higher precision but lower recall and F1-score than the U-Net ensemble. OA is also lower due to confusion between the unirrigated and uncultivated classes. The full U-Net has many more errors of omission which lead to underestimates of irrigated land, showing the utility of our method. The results in Table II are not entirely surprising, as the U-Net ensemble was trained exclusively on data from MT and uses a more powerful computer vision method than the other two. However, it does show that the U-Net ensemble is capable of classifying irrigated lands with higher precision and accuracy than other methods in MT, making the utility of the product in MT clear. All methods have an OA of greater than 96%, reflecting the imbalanced nature of the validation data. For other states, especially states where irrigated agriculture comprises a larger percentage of land cover, OA may be a better estimate of model performance. The low precision reported in Table II for LANID and IrrMapper RF is a result of both models misclassifying dryland agriculture as irrigated.

A. Validation With a Spatially and Temporally Distinct Test Set

As expected, the method generalizes even when tested against both spatially and temporally independent validation sets. Table III shows the F1-score of the model ensembles on the test set. Years (2008, 2013, and 2015) indicate that the set used to train the models did not include any data from the year specified. The F1-score reported is the ensemble's performance for that data from that year. Also shown is the performance of a single model chosen randomly from the ensemble. In each case, the ensemble of models does better than the chosen constituent.

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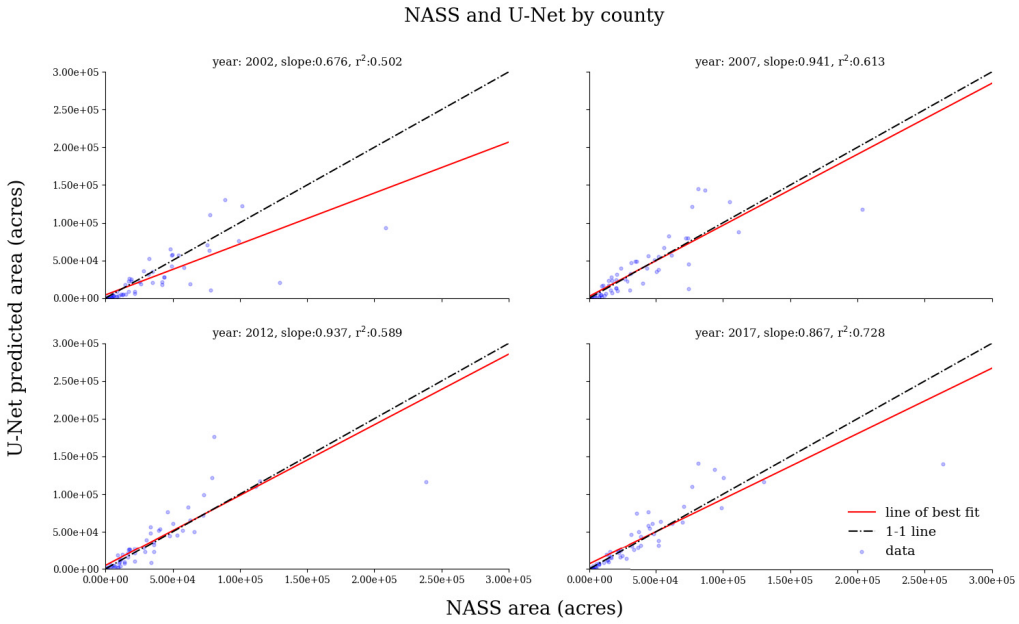


Fig. 3. U-Net ensemble results compared to NASS statistics. The red line indicates the line of best fit for the data, and the black line the 1-1 line.

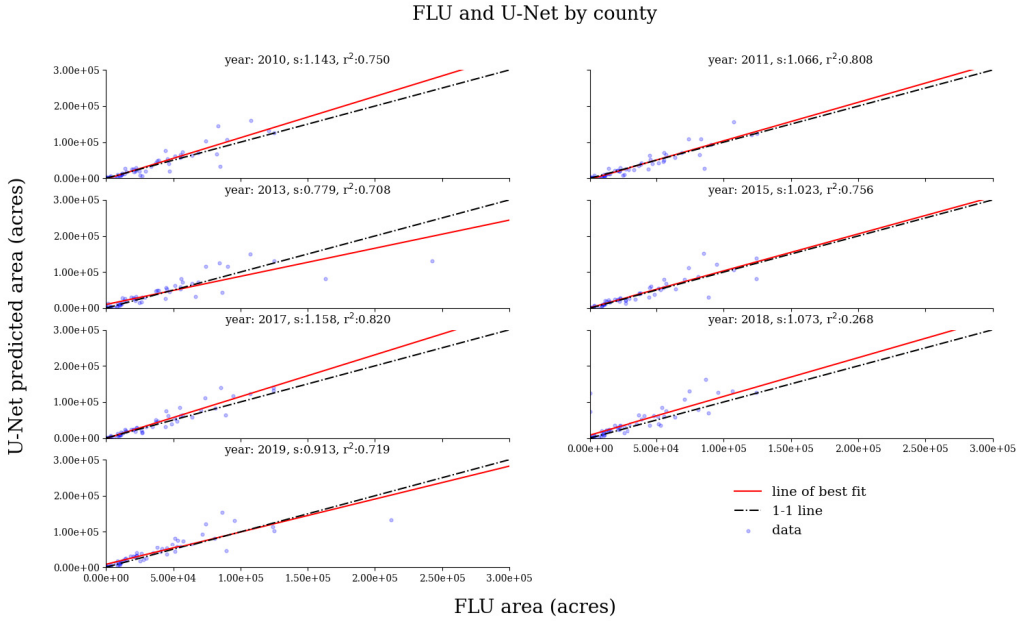


Fig. 4. U-Net ensemble results versus FLU statistics. The red line indicates the line of best fit for the data, and the black line the 1-1 line.

B. Validation With NASS and FLU

Here, we take a pixel-counting approach to total area computations, which we expect to be an unbiased estimator, because of our classifier's balance between precision and recall demonstrated in [32, Table 2]. Fig. 3 shows that the U-Net ensemble slightly underestimates irrigated area by county when compared to NASS. Fig. 4 indicates that the U-Net ensemble and FLU data agree fairly well.

The U-Net ensemble estimates more irrigated area compared to FLU data on a county-wide level for most of the years compared. Both comparisons show that variance in the comparison between census data (either FLU or NASS) increases with an increase in county irrigated areas. The total area of irrigated agriculture predicted by different methods is shown in Fig. 5. Overall, the median U-Net ensemble prediction is closer to both FLU and NASS than the two other methods examined (LANID and IrrMapper-RF). MIrAD-US agrees very well with NASS, as the method matches its predictions directly to NASS statistics.

There is a general increasing trend in irrigated area predictions in all methods of estimated irrigated area. IrrMapper-RF and U-Net ensembles agree on the trends in irrigated acreage

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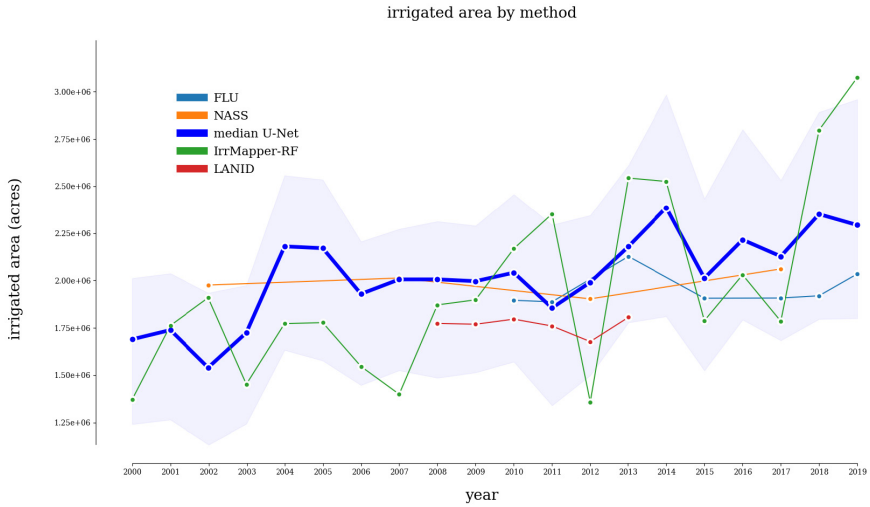


Fig. 5. Irrigated area in MT by method, with NASS and FLU data included. 25th and 75th percentiles for U-Net ensemble area predictions are shaded in blue.

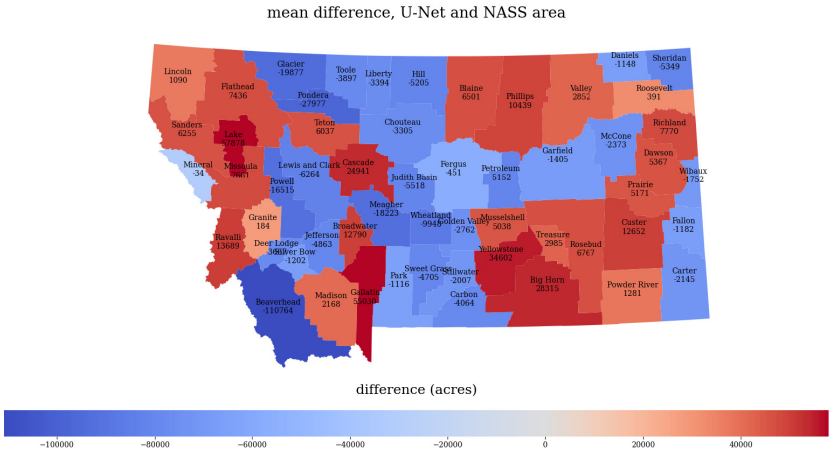


Fig. 6. Mean difference in U-Net ensemble estimates and NASS area by county. Mean taken over NASS census years.

that they identify. IrrMapper-RF predicted irrigated area follows trends in precipitation, most likely due to confusion in wet areas due to overall higher NDVI in wet years and possibly the detection of temporary increases in irrigated land because of increased water availability. The U-Net ensemble does not show the same correlation with precipitation as IrrMapper. In both comparisons to NASS and FLU, r^2 values vary widely, from 0.50 (2002) to 0.73 (2017). The large discrepancy between the U-Net ensemble and NASS area for certain counties in 2002 is attributable to a model error in counties on the Rocky Mountain Front. The model erroneously classified areas that consistently predict irrigated in other years as unirrigated. We found that clouds were present in every image save one, meaning the model had little to no information to make the classification. The clear outlier (bottom rightmost point) in all subplots in Fig. 3 is Beaverhead county. NASS reports irrigated area from 200 000 to 260 000 acres, while the U-Net ensemble reports area from 90 000 to 140 000 acres. FLU data agrees very well with U-Net ensemble results for Beaverhead county, with the smallest difference in irrigated area prediction at only 2000 acres. The U-Net ensemble consistently produces larger estimates of irrigation for a cluster of counties that NASS reports to have around 100 000 acres—Lake, Yellowstone, and Gallatin counties (Fig. 6). FLU estimates of area for the same counties agree closely with U-Net estimates of irrigated area (15 000-acre difference for 2017).

C. Qualitative Performance of Models

Inspection of Landsat and NAIP imagery for NASS census years of Beaverhead county indicates that the large discrepancy in the irrigated area between NASS and U-Net ensemble predictions is most likely due to U-Net ensemble omissions of flood irrigated lands as irrigated. The U-Net ensemble tends to classify flood irrigation or irrigation at the wetlands–agriculture interface as wetlands. It is often hard to tell where wetlands begin and irrigation ends when inspecting high-resolution satellite data, so it is not surprising that the U-Net ensemble has difficulty distinguishing these two classes. In addition, in our training data and our study area,

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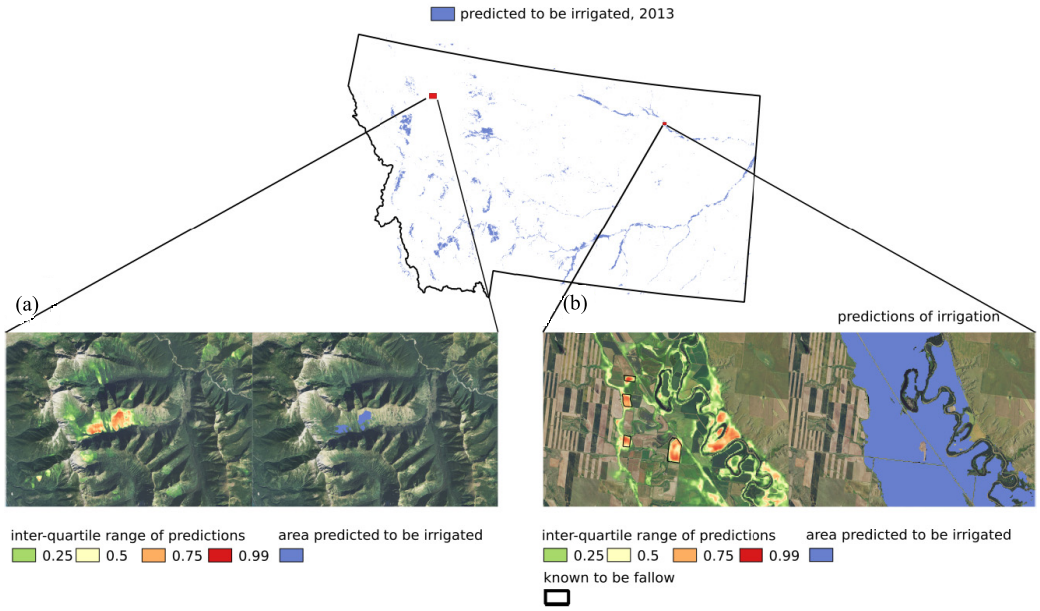


Fig. 7. Example of large IQR where the model has incorrectly classified forest and fallowed lands as irrigated. (a) Mountainous areas classified as irrigation (right), and the corresponding ensemble IQR (left). (b) Same phenomena as (a), but in a different region where the model ensemble incorrectly classified fallowed land as irrigated.

the existence of wetlands and irrigation in the same place is possible, and therefore both semantic and physical distinction between irrigated land and wetlands is blurred. This problem may be resolved by more extensive vetting of wetlands training data.

Each model has different biases on wetlands. LANID-2012 restricts its classification of irrigated land to the land that is not wetlands, impervious surfaces, open water, and forests according to the 2012 CDL. Some flood irrigated land is classified as wetlands in the CDL, resulting in errors of omission. This is evident when inspecting NAIP imagery in the regions that the CDL classifies as wetlands. IrrMapper-RF, on the other hand, tends to label irrigated land that LANID and the U-Net ensemble miss as irrigated. However, IrrMapper-RF also tends to label true wetlands as irrigated, for example, clearly uncultivated land on islands in rivers. The U-Net ensemble seems to bridge the gap between the two methods; it labels true wetlands as wetlands and some flood irrigated land as irrigated. However, the U-Net ensemble does not detect as much flood irrigated land as IrrMapper-RF, resulting in lower estimates of irrigation in counties such as Beaverhead. Some wetlands training data incorporate pixels where flood irrigated land was labeled as wetlands. It is conceivable that the labels were correct at the time of their creation but the land use changed during subsequent years. In general, we expect IrrMapper-RF to overestimate irrigated land near wetlands–agriculture interfaces, U-Net ensemble to underestimate somewhat, and LANID to almost entirely ignore. This is evident in the Big Hole Valley near Wisdom, MT, where LANID omits irrigated land. Classification of cultivated land at the wetlands–agriculture interface is difficult due to the possible semantic and physical overlap between irrigation and wetlands. Other land classes with less label uncertainty are accurately classified by the U-Net ensemble compared to other methods.

Dryland agriculture is very well classified by the U-Net ensemble compared to other methods, with few errors of commission. The errors mostly happen at the edges of irrigated fields that are labeled as dryland due to the difference in label and data resolution. LANID has many errors of commission in dryland areas, particularly in the eastern half of MT. IrrMapper-RF also overpredicts irrigation in dryland agriculture but has fewer errors of commission in dryland areas when compared to LANID. The accuracy with which the U-Net ensemble distinguishes irrigated and rainfed agriculture are one of the large advantages of using this product. The U-Net ensemble systematically overestimates irrigation compared to NASS for Lake, Gallatin, and Yellowstone counties. These are all counties with large areas of irrigated land that reside next to bodies of water, and as such have large irrigation schemes. U-Net also confidently misclassifies some fallowed fields. Fallow fields were not well represented in the training data compared to irrigated land. U-Net ensemble misclassifications in these areas could be solved by the creation of a larger fallow training set.

D. Uncertainty in Model Predictions From Prediction IQR

The models in the ensemble did not always predict similar probabilities for each pixel. Histograms of the IQR for misclassified pixels are shown in Fig. 8 (shown in the supplementary). The distributions of IQR for errors of commission on the irrigated class (whether on fallowed land, wetlands, dryland agriculture, or forest) are much different than the distributions for errors of commission on other classes. In particular, the pixels that are known to be nonirrigated yet predicted as irrigated tend to have distributions of IQR that peak in the

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0.4–0.8 range. The ability of the model ensemble to indicate some uncertainty makes directly fitting NASS or FLU data estimates of the area by county possible by considering pixels irrigated that are above or below some IQR threshold. Land near the wetlands–agriculture interface often has a higher IQR than land that is clearly irrigated, indicating uncertainty in classification. Similarly, some fallow land classified as irrigated has a high IQR. The model ensemble also exhibits high IQR in mountainous regions of northwestern MT that are erroneously classified as irrigated.

The histograms of IQR for all pixels that the models predicted as a certain class are shown in Fig. 9 (shown in the supplementary). Pixels predicted to be irrigated have a larger median IQR than pixels predicted as other classes. Even removing the first 30 bins for the IQR of the unirrigated and uncultivated class results in a median IQR of 0.02 and 0.03 for the unirrigated and uncultivated class, respectively. Medians without removal of bins for both classes are 0. Median IQR for the irrigated class was 0.1. Examples of misclassifications that are highly uncertain are shown in Fig. 7.

U-Net predicts irrigation in some small regions in the mountains of northwest MT. The IQR associated with the predictions clearly indicates that the ensemble is uncertain about the classification. Fig. 7(b) shows similar results for land that is known to be fallow—the fields outlined in black (which are known to be fallow but labeled by the ensemble as irrigated) clearly have a higher IQR.

The ten models in the ensemble perform differently because they were trained on a different subset of the training data. In general, models that performed poorly on the validation set tended to have a less "strict" definition of irrigation and correctly predicted irrigation at the wetlands–agriculture interface more frequently than the median prediction. In contrast, models that performed well on the validation set tended to have a more restrictive definition of irrigation and correctly identified more fallowed or dryland fields as irrigated in counties like Lake or Yellowstone. Part of this trend may be explained by the sparsity of labels in the validation set, as models that learn a very restrictive definition of irrigation may not truly reflect reality despite performing well on the validation set. All rasters used in the final analysis are available on GEE, and we recommend examining predictions qualitatively and examining model uncertainty when performing analysis.

IV. Conclusion

We present a new methodology of mapping irrigation (IrrMapper U-Net) at high spatial resolution and over large regions in time and space, applied to the state of MT. The method used is a deep learning model that ingests temporal mean images of satellite-derived spectral data from multiple times throughout the year. The model is also computationally efficient enough to produce predictions for 20 years over the state of MT on a personal computer in less than a day. Our method agrees better statewide with both NASS and FLU data than existing maps of irrigation in MT and outperforms other methods on a hand-labeled validation set in overall accuracy, F1-score, and precision. The misclassifications of our method come from underpredicting irrigation at the wetlands–agriculture interface and overpredicting irrigation in fallow fields. Despite these drawbacks, the model results offer classification accuracy improvements over existing methods, especially in rainfed areas, where there are almost no errors of commission. The uncertainty of our model ensemble is high in regions with high rates of misclassification, meaning thresholding final irrigation maps based on ensemble uncertainty is tenable. The method we propose is the first application of CNNs to mapping irrigation on a large spatial and temporal scale and was able to be applied with modest computational infrastructure.

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